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The brain of the Malayan bear (*Helarctos malayanus*)

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Abstract

This little known ursid has a brain that shows very progressive encephalization and neocorticalization, reminiscent of the giant panda's. But, somewhat paradoxically, it is also characterized by proportionately large rhinencephalon and brain stem, and a relatively small striatum. This quantitative study suggests a general contrast between Asian and American bears. It invites further comparative investigations of the brain in all ursiform mammals, as an especially promising material from combined evolutionary and neuroethological standpoints.

Key words: Carnivore brain – Ursidae – Malayan bear – Neuromorphology – Brain component

Introduction

Brains of bear-like mammals have been analyzed quantitatively in two recent papers. The giant panda (*Ailuropoda melanoleuca*) was compared with the North-American raccoon (*Procyon lotor*) and the black bear (*Ursus americanus*) by PIRLOT and JIAO (1985). The lesser panda (*Ailurus fulgens*) was examined by KAMIYA and PIRLOT (1988). In both articles, reference was made to the long-running debate about the exact taxonomic position of the two pandas with respect to the so-called "true" bears. We will not repeat the arguments here but merely point out that: a. various morphological criteria have been used over the past forty years for and against an ursid position for the giant panda; b. recently, immunological and molecular approaches seem to have decided in favour of an ursid position (O'BRIEN et al. 1985 and a short review in KAMIYA and PIRLOT 1988).

This paper focusses on a true ursid, the Malayan bear *Helarctos malayanus*, and compares it with two other true ursids, that is, the black bear and the giant or greater panda. Occasional reference will be made to the lesser panda and raccoon for the sake of morphological comparisons.

Materials and methods

Dr. M. SAITOH, veterinary surgeon at the Tama Zoo, near Tokyo, kindly supplied the brain of a male *H. malayanus*, one of several conspecific animals exhibited in the zoo as "Malayan bears" or "sun-bears". The species is known to inhabit the hot tropical forests of Southeast Asia, including some Indonesian islands, and it is described as a good climber (MORRIS 1965).

The specimen was reported to weigh 82 kg at the time of death. According to MORRIS (1965), however, the Malayan bear is rather small for an ursid: the author mentions a weight of 100 pounds. In JERISON (1973, p. 359), 45 kg is given for the species: this weight may have been taken from the same source as MORRIS', but this is not specified. Nor is there any reference for JERISON's figure of 390 g as brain weight, which is a little higher than the 300 g weight for our specimen. If the Malayan bear is as small (around 45 kg) as reported by those authors, its relative brain size is well above the range established for all other ursids, or even for carnivores in general; it is, in fact, even higher than in the majority of primates and a number of dolphins. We thus sought and obtained confirmation of the figures for our *Helarctos* individual. This question will be discussed below.

Progression indices were used to estimate the relative size of total brain and of brain parts, a method that has been explained several times (for example, in PIRLOT and KAMIYA 1982). The index expresses how many times the brain or a brain part in *Helarctos* is larger (or smaller), relative to body, than those to be found in a theoretical and primitive mammal of equal body size. An index above 100 indicates a progression in relative size with respect to the reference type.

The external features of the brain of *Helarctos* were considered from the dorsal, lateral and ventral aspects.

Results and discussion

A. Surface morphology

The Plate and Fig. 1 show the external morphology of the brain of *Helarctos* compared with that observed in other ursiform mammals. Many features could be used for this comparison but our remarks will be confined to a laterodorsal area which is known to have important functional significance in mammals and has been described several times in the past, albeit generally not from a comparative viewpoint.

1. The following superficial traits, among others, can be noted. The brain of the Malayan bear is well proportioned in that the cerebrum is almost "square" in polar view (90 mm in length and 85 mm in breadth, in fresh condition – see Plate). It is also fairly bulky in lateral aspect with a height of 65 mm. The olfactory bulb, olfactory stalk and piriform area are large and clearly protruding. The cerebellum is well developed; its transverse width was again 65 mm in the fresh specimen.

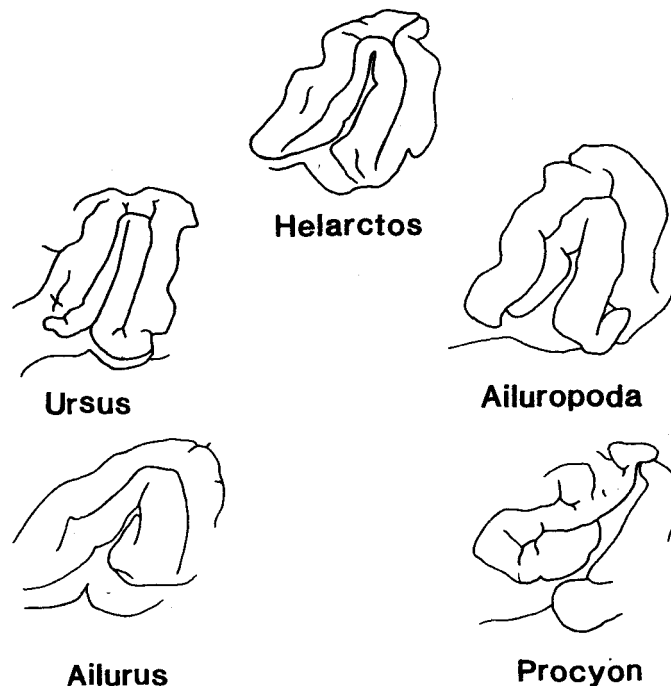


Fig. 1. The Sylvian (pseudosylvian) area in five ursiforms. To understand these figures, one must refer to Fig. 2. 1. First note that gyri RS and CS, present in the dog, do not exist in the bearlike species; they are shown as land-marks only. Then, by comparing, one can see that the two gyri bordering the Sylvian sulcus (10) in these mammals are those numbered 8 (rostral ectosylvian – left) and 9 (caudal ectosylvian – right) in the dog. The variations in the positions of these two gyri are commented upon in the text (A. Surface morphology)

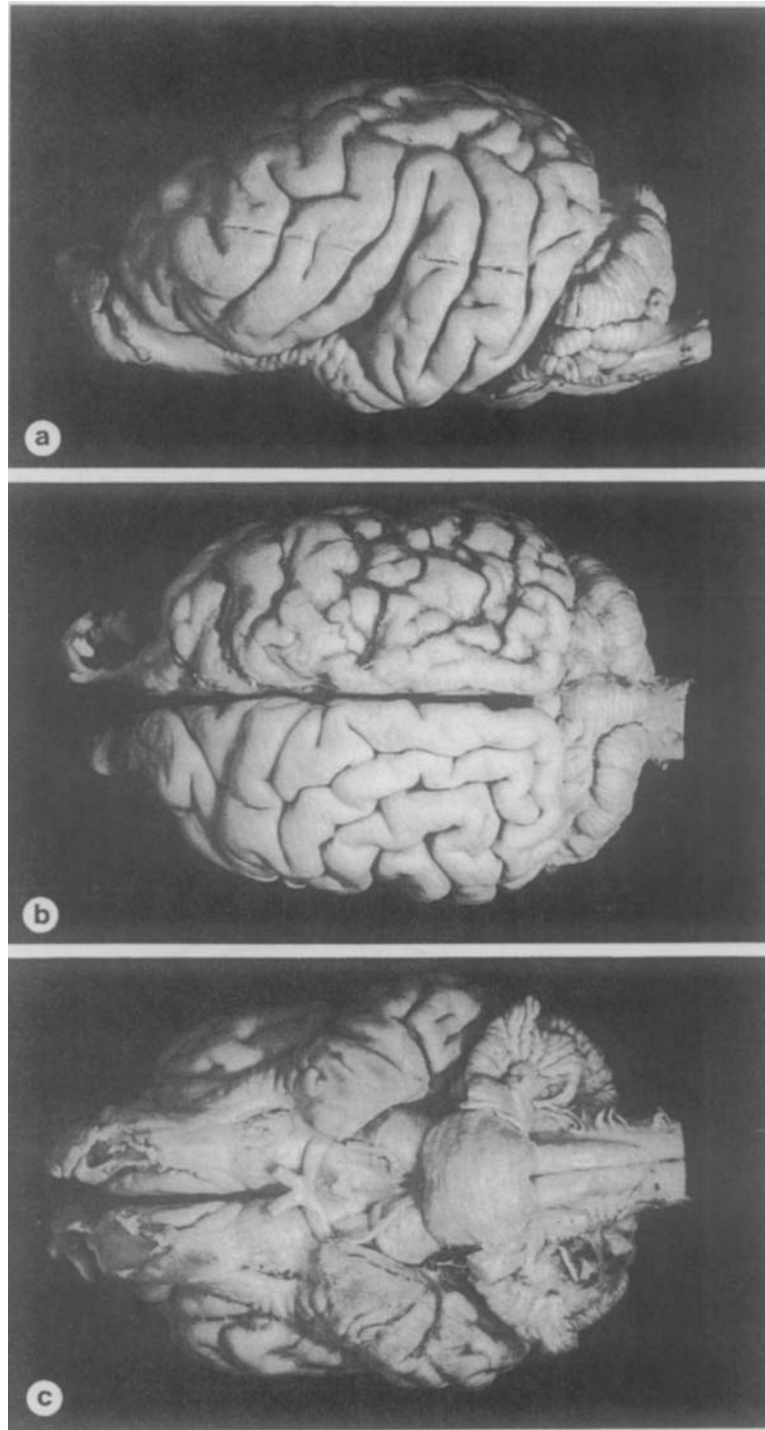


Plate. External aspects of the brain of *Helarctos malayanus*. a: lateral; b: dorsal; c: ventral

2. Looking at the gyri from the laterodorsal side, one can see that the whole lateral pattern is very similar to that in the black bear (Fig. 1). There are, however, some differences between the two animals with respect to this anatomical region, as well as certain dissimilarities between the "true" or "traditional" bears and the other species (pandas and raccoon).

To start this comparison with a familiar and classical pattern, one can take as a reference model the laterodorsal aspect of the cerebrum in the dog, as presented in various textbooks, for example in JENKINS (1978) (Fig. 2). Of particular relevance here are the following sulci and gyri (Figs. 1 and 2): 1. cruciate sulcus; 2. precruciate sulcus; 3. postcruciate sulcus; 4. precruciate gyrus; 5. postcruciate gyrus; 6. coronal sulcus; 7. coronal gyrus; 8. rostral ectosylvian gyrus; 9. caudal ectosylvian gyrus; 10. Sylvian sulcus, better known as pseudosylvian sulcus in the ursids – the terminology adopted here. Two important gyri appearing in the dog but not seen on the lateral surface of the ursiform cerebrum are the rostral and caudal Sylvian (Fig. 2–RS and CS).

It is then possible to label, in the five mammals here studied, some of the gyri and sulci mentioned above (see Fig. 1). The first observation is that the pseudosylvian sulcus (10) is more or less open. In the Malayan bear, it is a rather widely open cleft; in the black bear, the rostral and the caudal ectosylvian gyri are in contact with each other, thus virtually closing the cleft, as can be seen in PAPEZ (1967, Fig. 22, p. 24), although a previous sketch by PIRLOT *et al.* (1985) gave the impression that it was slightly open; in the giant panda, most of the sulcus is open, although its dorsal segment is closed by the apposition of the two bordering gyri; in the lesser panda, the ectosylvian gyrus is well exposed and the pseudosylvian sulcus remains open, as in the giant type, over most of its length, but, in its dorsal segment, there is again an apposition of the rostral on the caudal gyri; in the raccoon, as reported by PIRLOT and JIAO (1985) and PIRLOT *et al.* (1985), the rostral ectosylvian gyrus is visible ventrally, while its dorsal portion lies hidden in the depth of the pseudosylvian fissure, and the two ectosylvian gyri do not make contact ventrally, so that the sulcus, here too, is partly open.

This pattern sequence represents various extents of opercular formation (see PAPEZ 1967, p. 25). Such an "opercularization" is brought about by a downward and caudal expansion of the coronal gyrus, while the precrucial and postcrucial systems develop at the same time in a downward and rostral direction.

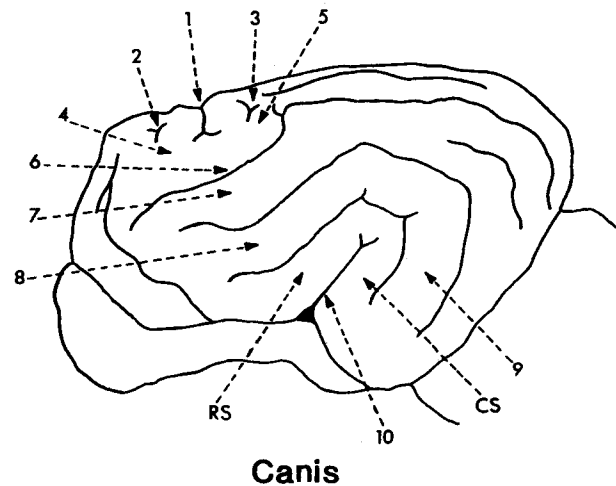


Fig. 2. Lateral aspect of a dog's brain. 1 = Cruciate sulcus; 2 = Precruciate sulcus; 3 = Postcruciate sulcus; 4 = Precruciate gyrus; 5 = Postcruciate gyrus; 6 = Coronal sulcus; 7 = Coronal gyrus; 8 = Rostral ectosylvian gyrus; 9 = Caudal ectosylvian gyrus; 10 = Sylvian or pseudosylvian sulcus; RS = rostral Sylvian gyrus; CS = caudal Sylvian gyrus

From a functional standpoint, the difference between ursiforms and dog can be interpreted as a trend toward an oversizing of two areas of great significance in the ursiforms: a. the motor area, which may be especially responsible for the skillful use of the hands and fingers; b. the coronal gyrus (7), which is known to be a sensory area, and may in turn control the sensitivity of the extremities of the limbs (see MAGALHAES-CASTRO and SARAIVA 1971; HERRON 1978; SARAIVA and MAGALHAES-CASTRO 1975). The increment of this area of the brain in the five ursiforms would thus represent an early stage of the opercularization process that reaches completion in higher mammals.

These remarks are a tentative functional interpretation of the surface anatomy of the main laterodorsal area of the brain in these mammals. It is clear that no definite statement can be made until more data are available from physiological investigations using probing electrodes. Then only will it be possible to make an accurate estimate of the size of each specific area devoted to each specific function (motor or sensory) in every one of the five species. Obviously, the thickness of the gyri must also be considered if one wants to arrive at an integrated interpretation of these structures, that is, one in which qualitative and quantitative aspects are combined.

B. Volumetry

Two main points will be considered: 1. the relative size of total brain; 2. the relative size of its most important components.

Total brain

The size of total brain relative to body is "encephalization". Expressed with reference to the ancestral mammal (progression index), its value in the Malayan bear is 564 with a body weight of 82 kg. In comparison, an extremely progressive giant panda had an index of 692 (PIRLOT and JIAO 1985), while a group of four males and two females yielded an average figure of 585 (PIRLOT et al. 1985). At first sight, the index 564 for the Malayan bear does not appear exceptional. However, using the data from JERISON (1973 – body 45 kg and brain 390 g), we previously obtained an index of 1065 for the same species (PIRLOT et al. 1985), that is, almost twice as high as that in this study. With a body weight of 45 kg or 100 pounds (instead of the "fat" 82 kg) for our specimen and 300 g for its brain (in agreement with both MORRIS 1965, and JERISON 1973), the progression index would rise to 820. This figure too seems high and thus should be viewed with caution; it is not very much lower than the figure 1065 mentioned above. Table 1 in PIRLOT et al. (1985) in which indices are given for 18 species of carnivores, shows a range from 277 to 631, then a jump to 1065, which does not inspire confidence. Many primates, including several catarrhines, have index levels below 800, and PIRLOT and KAMIYA (1975) found a few cetaceans with indices below 700 along with some exceptionally encephalized species.

We therefore tend to believe that, although in the upper range of the index values for carnivores, the Malayan bear is unlikely to be much more encephalized than the giant panda, which we here consider a true ursid. In all likelihood, these animals are at about the same level, that is, within or not very much above the 537–692 range (PIRLOT et al. 1985, Table 2).

The technical difficulty we experienced with our specimen confirms the relative unreliability of body weight in mammals. This problem has been mentioned before (PIRLOT 1981, 1987). In the present case, we were obviously dealing with two "abnormal" individuals: a Malayan bear that had become overfat from zoo diet, and a greater panda that had died after a long illness and clearly was underweight for its age (PIRLOT and JIAO 1985; PIRLOT et al. 1985). Hence, the former yielded indices that were too low and the latter, too high figures. This does not preclude reasonably valid conclusions, but only thanks to additional information available from the percentile composition of the brain, and ultimately by drawing from experience with quantitative morphology of the mammalian brain.

Brain components

This discussion of brain components will be confined to the three ursiforms mentioned as true ursids, that is, *Helarctos*, *Ursus*, *Ailuropoda*. Three sets of figures on the relative size of components will be considered.

Progression indices

The progression indices for the Malayan bear are presented together with those for the black bear and the giant panda (Table 1). The following remarks may be made, using the abbreviations in the table.

BO: The progression of the bulb is strikingly similar in all three animals (31, 38, 38). This may point to a great similarity in olfactory perception of the environment. Since the panda is almost exclusively, and the other two bears are partly vegetarian mammals, it may be that all mainly use olfaction in relation to a fruit-chlorophyll diet, not excluding, however, grub and many small vertebrates, at least in the case of *Helarctos* and *Ursus*. Of course, phylogenetic progression of bulb alone does not necessarily indicate the type of food preferred, and, among these three related carnivores, olfactory sensitivity is almost surely greater in the American than in the Asian species, as suggested by the proportions in Table 2.

N: The large interval previously found (PIRLOT and JIAO 1985) in neocortical progression between panda (3018) and black bear (1635) is occupied about mid-way by our Malayan bear (2247). Assuming that the figure for our specimen is too low, the two Asian species should be fairly close to one another and definitely higher than the American form.

RH: The black bear has a very exceptional figure for the rhinencephalon, 96, which means regression with respect to the basal mammalian type (or at least absence of significant progression, if one takes into account the fact that the measurements and calculations are approximate and accept that 96 % does not really differ from 100). The other two animals are moderately progressive, that is, they yield indices 140 (panda) and 195 (Malayan bear) respectively. The occurrence of a regressive along with two progressive figures is difficult to explain. The relatively high progression in *Helarctos* is also unexpected. It is possible that some non-olfactory function became enhanced in the RH area of Malayan bear, resulting in a local enlargement of the component in this species in contrast with the others and, perhaps, this occurred in the Malayan line alone. This is merely speculative but, in connection with this idea, it might be worth investigating the functions and volumes of the limbic elements present in the concerned area (amygdala).

ST: The range of progression of the striatum is very broad with an exceptionally high value in the panda (1027) and a low one in the Malayan bear (601). These figures point to distinct phylogenetic evolutions for this component in the three bears. One may wonder if the high index for ST in the panda could be a conservative feature. On the other hand, from a functional and adaptational viewpoint, the indices give no information, but the proportions do suggest a more important role for the ST in the black bear and a less important role in the Malayan species (Table 2). Does this relate to some behavioural difference in motor control (including the fine use of anterior extremities)? This interesting question cannot be discussed here, but it will hopefully be investigated in a later work in the perspective presented by PARENT (1986) in his review of the mammalian basal ganglia (chap. 7 to 10).

H: The hippocampus is less progressive in the Malayan and black bears than in the panda. Again, we do not know if this has to do with some evolutionary particularity of the limbic system. More aggressive animals perhaps tend to exhibit a less progressive H in their evolutionary line (see STEPHAN and PIRLOT 1970). The panda has the highest progression index and also the highest encephalic proportions for H (Table 2). But to make a final judgment on the functional significance of these figures requires more comparative behavioural data.

Table 1. Progression indices

	<i>Helarctos</i>	<i>Ailuropoda</i>	<i>Ursus</i>
BO	31	38	38
N	2247	3018	1635
RH	195	140	96
S	144	162	136
D	415	478	258
ST	601	1027	641
H	197	259	116
M	461	311	227
C	510	613	317
O	275	136	93

Abbrev.: BO = olfactory bulb; N = neocortex; RH = rhiencephalon; S = septum; D = dien-cephalon; ST = striatum; H = hippocampus; M = mesencephalon; C = cerebellum; O = medulla oblongata

C: If the progression of the cerebellum is directly related to motor skills (walking, climbing, jumping and especially handling of objects), then, one should not be surprised to see a high C value in the panda and a fairly high figure in the Malayan bear too. The relatively low C value in the black bear, however, is puzzling. To our knowledge, there are no detailed comparative studies on the climbing performances or the handling of plants or animal preys by the three animals.

O: The O value is unexpectedly high in the Malayan bear and extremely low in the black bear, almost suggesting an evolutionary divergence among ursids. Although all indices in the black bear point to a relatively low general progression, a regressive (or non-progressive) figure was hardly expected; this agrees with the above remark on the low index (96) for RH in the same species. The encephalic proportions of the three animals, however, are not unexpected (Table 2) and agree with the high index in the Malayan bear.

It can be said that, in general, the brain components in the Malayan bear show quantitative progressions which fit well into the ursid framework of brain evolution. They are often intermediate between the panda's high progression and the black bear's low progression, and are probably closer to the panda's. However, progression indices merely describe changes in brain size relative to body size in a phylogenetic perspective, that is, with respect to an arbitrary ancestral reference type. Functional interpretations of the indices are often open to question (see PIRLOT 1987 and in press). A brief consideration of the development of parts within the brain itself may in part offset this drawback.

Table 2. Percentile composition of the brain in the three ursids
(Abbrev. see Table 1)

	Of total brain			Of telencephalon		
	<i>Helarctos</i>	<i>Ailuropoda</i>	<i>Ursus</i>	<i>Helarctos</i>	<i>Ailuropoda</i>	<i>Ursus</i>
BO	0.62	0.61	1.12	0.87	0.80	1.45
N	59.47	64.42	63.00	82.97	84.07	82.13
RH	5.10	3.08	3.73	7.11	4.01	4.85
S	0.31	0.30	0.44	0.43	0.39	0.58
D	5.23	4.99	4.82	—	—	—
ST	4.21	5.99	6.69	5.88	7.82	8.72
H	1.96	2.21	1.74	2.74	2.89	2.27
M	3.22	1.87	2.41	—	—	—
C	15.44	14.64	13.78	—	—	—
O	4.44	1.89	2.27	—	—	—
	100.00	100.00	100.00	100.00	100.00	100.00

Intra-encephalic proportions

Size ratios of brain components to total brain and to telencephalon for the three ursids are presented in this section (Table 2).

In the percentile composition within the total brain, none of the figures shows any major departure from the general trend. The most interesting observation is that the Malayan bear has the lowest percentage (59.47%) for the neocortex and the highest for the rhinencephalon (5.10%). The striatum of this species is clearly lower (4.21%) than that of the panda (5.99%) and of the black bear (6.69%). Its cerebellum is also relatively the largest, but not to any great extent. Its hippocampus is not very different in encephalic proportion from those of the panda and the black bear.

The rostral part of the brain (telencephalon) can be compared to its caudal section (brain stem in many authors, that is diencephalon plus mesencephalon plus medulla oblongata) which may be taken as representing the initial neural tube in the head without the forward outgrowth. The telencephalon/brain stem ratios so defined are 5.56 in *Helarctos*, 8.92 in *Ailuropoda* and 8.19 in *Ursus*. Table 2 shows that the three caudal components (D, M and O) are relatively larger in the Malayan bear than in the two other bears. Taking a more restricted definition of the brain stem, that is M+O only, we also find that this combination is largest in the Malayan bear (7.66 in *Helarctos*, 3.76 in *Ailuropoda* and 4.68 in *Ursus*). Using these figures for a telencephalon/brain stem ratio, we obtain: 9.36 for *Helarctos*, 20.37 for *Ailuropoda* and 16.39 for *Ursus*. The large size of the back-brain in the Malayan bear is thus exceptional.

Regarding the percentile composition within the telencephalon, almost identical observations can be made. There is again a high proportion of RH in the Malayan bear (7.10%) while the panda reaches only 4.01% and the black bear 4.85%. The percentages for the striatum correspond to the progression indices (5.88% in the Malayan bear, 7.82% in the panda and 8.72% in the black bear). These observations, which were to some extent expected, in agreement with the more general data on total brain percentages given above, reinforce the previous conclusion that the neocortex is about equally developed in all three animals, while the rhinencephalon is proportionately somewhat larger and the striatum a little smaller in the Malayan bear than in the other two ursids.

Conclusions

The morphology of the brain of the Malayan bear in comparison with the two other ursids can be viewed in more than one way.

1. The surface anatomy is very similar in all three animals. This is especially true for the lateral aspect of the brain (temporal and parietal regions). However, there are enough small differences in the perisylvian area to suggest that particular functional features would probably be observed if detailed electrophysiological mapping of the lateral gyri were feasible.
2. The size data on total brain and brain parts raise interesting questions about to the evolutionary history of the three animals, especially the Malayan bear. The relatively high encephalization of *Helarctos* may be partly and simply explained by the fact that this species is a rather small form for a living bears, while the high figures obtained for *Ailuropoda* by PIRLOT and JIAO (1985) can probably be accounted for to some extent by the fact that the individual they studied was sick. On the other hand, *Helarctos* happens to be a tropical animal, dwelling in hot humid forests, as opposed to the other two species which are encountered in colder regions, sometimes enduring severe winters, and exhibit larger body sizes, perhaps in agreement with Bergmann's rule.
3. The volumetry of the brain components in the Malayan bear is intriguing. The progression indices (which, it must be remembered, entail a phylogenetic approach – Table 1)

clearly show that the neocortex is more progressive in the two Asian than in the American species, especially if, as said before, the figure 2247 is probably an underestimate. It is also certain that the lower brain parts (D, M and O) are more progressive in the Malayan bear than in the other two. Although indicating a real contrast between eastern and western species, these figures do not relate to any known ecological or ethological difference. Whether this could be related to a difference in basal metabolic rate is a question that cannot be answered until physiological data are available (see discussion in PIRLOT 1987 and in press).

A better idea of the relative importance of the various functions subsumed by each encephalic component can be gained by looking at the percentile composition of the brain in each bear, at least in the case of those components that have been sufficiently investigated by neurophysiological methods. All three bears are at about the same level as far as neocortical size is concerned; this level is fairly high, corresponding to that found in certain simians; four platyrrhines and one catarrhinian, for example, possess neocortical ratios between 57 and 65 % of total brain (based on data in STEPHAN et al. 1970). With regard to the olfactory components taken together, the Malayan bear is highest, and it would be interesting to know more about its olfactory habits; its bulb, however, is less developed than the black bear's and the development of this component may be the best indication of the intensity of peripheral function. The striatum is smallest in *Helarctos* and largest in *Ursus*, and for this no explanation can be offered at the present time. The primary components mesencephalon and oblongata (that is, essentially, the brain stem) are considerably larger in the Malayan bear than in the others, and, for the time being, this remarkable feature too defies interpretation; the same is true of the group D, M and O, which can be taken as the "basic brain" too. Obviously, a great deal more must be known about the respective life habits of the three ursids before the quantitative composition of their brains can be understood in functional terms. The encephalic or telencephalic percentages, which are free of the drawbacks suffered by any reference to body size, confirm that the Asian bears are clearly distinct from the American, usually having better developed brain components but with one notable exception, the striatum, which is larger in the American species.

The observations reported here are but a start for a satisfactory study of ursid brain morphology. They invite similar work on the Polar and Andean bears, on the one hand, on the Tibetan and European genera, on the other. Ecological and ethological data should throw more light on both the evolutionary and functional characteristics of these brains. In this respect, bears appear to be an especially suitable research material because they are distributed world-wide without, however, any major overlap. They are large animals with a relatively long life span. They are often very catholic in their eating habits or are downright carnivorous. Such a setting strongly contrasts with that in which most small rodents, plentiful and cheap to obtain, have always lived, which made them so popular for evolutionary studies; yet, ursids are less forbidding in size than many other large mammals, and their brains are relatively easy to secure, from zoos or even from the wild, when they are not protected. Documenting their life habits, again either in captivity or in freedom, should not pose more problems than have been solved so successfully by many studies that have been conducted on large felids, ungulates and primates by ecologists and behaviour scientists. It may be that the distinctions between the bears' brains are related to differences in "general psychology" (social life included) rather than to detailed specializations in their sensory-motor systems. All in all, the living ursids offer a promising challenge to the evolutionary neurobiologists.

Note. 'Neuroethology versus molecular, immunological and genetic approaches.' This article was completed when the authors found the superb summary of the bear-versus-panda riddle, and its solution, in O'BRIEN (November 1987). One may wonder if there is conflict between the two kinds of approaches. There should be none, and one of us will touch upon this methodological issue else-

where (PIRLOT, in press). We wish to emphasize here that, although it exhibits some clear similarities with the panda and some clear dissimilarities from the black bear, in its encephalic traits, *Helarctos* is a member of the Ursinae subfamily, while the panda is now considered to represent by itself the special Ailuropodinae subfamily, both within the Ursidae family (O'BRIEN 1987, p. 107). Macrofeatures such as brain organisation and behaviours do not necessarily coincide with a dichotomous classification based on certain microcharacteristics such as chromosome morphology and immunological responses, which nevertheless are at the root of macrodifferentiations, as determined by genomes. Convergences and divergences in brains and behaviours may be brought about by environmental selection acting on corresponding genotypes and other microcharacteristics. As already remarked by PIRLOT (1987), neurobiologists are still far from being able to disentangle the minute microfeatures of brain components in the impressive manner biochemists analyze isoenzymes and antibodies. But they are nearing that capability with a combination of qualitative and quantitative methods at all levels, micro and macro.

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Zusammenfassung

Das Gehirn des Malayenbären

Das Gehirn dieses wenig bekannten Bären zeichnet sich durch eine sehr progressive Encephalisation und Neocorticalisation aus. Damit erinnert es an den großen Panda. Aber – in gewisser Hinsicht paradox – ist es auch durch ein verhältnismäßig großes Rhinencephalon und ein verhältnismäßig kleines Striatum zu kennzeichnen. Die quantitative Betrachtung weist auf einen allgemeineren Unterschied zwischen asiatischen und amerikanischen Bären hin. Weitere quantitative Studien der Gehirne aller ursiformen Säugetiere versprechen interessante Befunde, weil sich dieses Material für eine Verknüpfung phylogenetischer und neuroethologischer Betrachtungen eignet.

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